Key Switches

Some parameters of SWAM instruments can also be triggered by the use of key switches. If there is an overlap in the instrument range after transposition, the key switches can be moved down one octave using the "KS Octave" parameter (see Advanced section). Key switches can be disabled by turning "KS Octave" To "OFF".

Key switches are colored red on the instrument's virtual keyboard.

Important note about Key Switches: "latch" Key Switches are sensitive to the KS velocity, even for two-state Key Switches. This way a sequence can be played correctly even if it is not started from the beginning; otherwise a "latch" status could work differently from how it should.

"KS Velocity Remap" adjusts the distribution of the KeySwitch values

The Key Switches provided are:

- C = Play Mode (at next Note On)
  - Low Velocity = Bow
  - Mid Velocity = Pizzicato
  - High Velocity = Col Legno
- C# = Manual Bowing
  (see "Manual Bowing: KeySwitch" in Play Modes/Right Hand section)
  - Tremolo: Note-on / Note-off
  - Bow Change: Note-on only
- D = Gesture Mode (at next Note On)
  - Low Velocity = Expression
  - Mid Velocity = Bipolar
  - High Velocity = Bowing
- D# = Alternative Fingering (at next Note On)
  - Low Velocity = Mid Position
  - Mid Velocity = Near the Bridge

- High Velocity = Near the Nut + Open

- E = Bow Lift
  - Off String (default)
  - While this key is held = On String
- F = Bow Start
  - Down Bow
  - Up Bow
- F# = Harmonics
  - Off (default)
  - Low Velocity = 2nd harmonic
  - High Velocity = 3rd harmonic

  Note: it's not possible to control the 4th harmonic through Key Switches
- G = Keep Bow Direction
  - OFF (default)
  - While this key is held = ON
- G# = Tremolo
  - OFF (default)
  - Low Velocity = Slow
  - High Velocity = Fast
- A = Tremolo Mode (at next Note On)
  - Low Velocity = Hz
  - Mid Velocity = Sync
  - High Velocity = Sync/Acc
- A# = Sordino (at next Note On)
  - Low Velocity = OFF

- High Velocity = ON
- B = 2nd page

2nd page of Key Switches (hold B Key Switch)

- B+C: Bow Real Mono
- B+C#: Bow Mono
- B+D: Bow Double
- B+D#: Bow Double/Hold
- B+E: Bow Auto
- B+F: Pizz/Col Legno Mono
- B+F#: Preferred strings 4-3 (for Double/Hold only)
- B+G: Pizz/Col Legno Poly
- B+G#: Preferred strings 3-2 (for Double/Hold only)
- B+A#: Preferred strings 2-1 (for Double/Hold only)

NOTE: All parameters controlled by the Key Switches can be controlled by MIDI Control Change, Aftertouch and NRPN messages as well, through the Controller Mapping section.

Instruments ranges

	Middle C = C3		Middle C = C4	
	Additional range	Regular range	Additional range	Regular range
SWAM Violin	D#2 - F#2	G2 - F6	D#3 - F#3	G3 - F7
SWAM Viola	G#1 - B1	C2 - C6	G#2 - B2	C3 - C7
SWAM Cello	F0 - B0	C1 - F5	F1 - B1	C2 - F6
SWAM Double Bass	B-1 - D#0	E0 - G#4	B0 - D#1	E1 - G#5

Notes:

- Ranges are provided in Concert (or Natural) pitch.
- Middle C can be switched from C3 to C4 in Main Menu -> Settings -> Options.
- To extend the lower range of the instrument act on the "String tuning" parameter in the "Advanced" page.
- SWAM Cello and SWAM Double Bass are transposed -12 semitones by default. Set the Transpose parameter to 0 to play with real pitch.

How to perform the main articulations

Almost all articulations are performed by the combination of the three principal controls: Note On velocity, expression, and bow-pressure. The instrument is so versatile that multiple articulations can be combined together - for example, tremolo/vibrato, tremolo/glissando, glissando/vibrato, glissando/harmonics/crescendo are possible. Here's a short list of main articulations.

Détaché

Detaché articulations are performed by separating the notes while pressing the Sustain pedal: the Note Off of the first note must happen before the Note On of the second note. A Slurred Legato is performed when the notes are overlapped while pressing the Sustain pedal.

Martelé

Set the "Bow Lift" parameter to "On String" and use high velocities and high expression at Note-on, then decrease the expression.

If you want to obtain a scratchier attack, set the "BowPressure" to a high value (0.65 – 0.80) just before the attack. Then, quickly decrease it to the desired value while decreasing the expression.

Spiccato

Set the "Bow Lift" parameter to "Off String" and play short staccato notes.

Legato

Slurred legato

A pure "slurred" legato is performed by overlapping the second note to the first, using a high value of Note On velocity for the second note, when the second note is selected on the same string as the first one.

Cross-string legato

A "cross" legato is performed by overlapping the second note to the first, using a high value of Note On velocity for the second note, when the second note is selected on a different string than the first one.

The choice Slurred vs Cross-string legato is mainly determined by the "Alternate Fingering" parameter. Setting the "Alternate Fingering" parameter BEFORE pressing a note determines which string is selected for the pressed note.

The actual threshold between portamento (see next paragraph) and legato is influenced by the "Portamento Max Time" parameter.

The legato transition "quality" is influenced by the "Dynamic Transitions" parameter.

Portamento (glissando)

A glissando between two notes is obtained by overlapping the second note to the first, using a low value of Note On velocity for the second note. If the "Portamento Control" parameter is set to "CC" just overlap the second note after setting the "Portamento Control" CC value to a low value (depending on the MIDI mapping curve defined for the "Portamento Time" parameter - See "Controller Mapping" section).

The actual portamento time and the threshold between portamento and legato are influenced by the "Portamento Max Time" parameter.

Continuous vs Split Portamento

If the second note is selected on the same string, a continuous portamento on the same string is performed. Otherwise, a "split" portamento across two strings is performed starting from the string of the first note to the string of the second note. The "split point" of the "split" portamento is determined by the "PortamSplit Ratio" parameter.

How to obtain a portamento on the same string

To obtain a wide portamento on the same string, avoiding a split portamento across two strings, you need to control the starting and ending position of the finger on the fingerboard acting on the "Alternate Fingering" parameter.

To do this in real time, use the "Alternate Fingering" Key Switch D# or control the "Alternate Fingering" parameter through a MIDI CC (see "Controller Mapping" section).

Note that the effective finger position changes at the next Note On, so you have to be able to set the right finger position BEFORE pressing the portamento starting note and also BEFORE pressing the portamento ending note.

Example:

1. Set the "Alternate Fingering" position to "Nut+Open" (K.S. D# - high velocity, MIDI CC – high value).
2. Press the note E3, it should perform on the string D, near the nut.

3. Set the “Alternate Fingering” position to “Bridge” (K.S. D# - mid velocity, MIDI CC – mid value).

4. Press the note E4, it should perform on the same string D, near the bow.

Flautando

To get a “flautato” sound, set the “Bow Pressure” parameter to a very low value. The text “Flautando” is shown near the Bow Pressure slider on the main GUI.

Scratch

To obtain a very scratchy sound, set the “Bow Pressure” parameter to the maximum value and play with high expression. The text “Scratch” is shown near the Bow Pressure slider on the main GUI.

Tremolo

Auto-tremolo

This selects if you want a “Slow” tremolo or a “Fast” tremolo. If the “Tremolo Mode” is set to “Sync” or “Sync/Acc”, the tremolo will be synchronized with the project tempo (BPM). Custom accents can be performed by acting on the expression (make expression “spikes” on the desired strokes).

Manual-tremolo

- Using Key Switch C#: be sure that Key Switches are active (Advanced / MIDI section). Set the “Manual Bowing: KeySwitch” to “Tremolo” (Play Modes / Right Hand section); press and release the Key Switch C# while playing the notes; a bow-change is performed at both Note On and Note Off.

- Using the currently playing note: set the “Manual Bowing: Sustain+Note” parameter to “Tremolo”; hold the Sustain pedal, press and release the currently playing note; a bow-change is performed at both Note On and Note Off.

You can adjust the “Bow Pressure” parameter in order to obtain a smooth or hard tremolo.

Bowing

Set the Gesture Mode to Bowing and move your expression controller back and forth.

Adjust the “Bowing Sensitivity” parameter to modify the amount of bow speed in function on how fast the controller speed varies.

Crescendo

Standard Crescendo: Crescendo is performed acting only on the Expression, starting from a low value (or from “niente”) and increasing it as desired.

Wider Crescendo: A wider effect can be obtained by mapping the “Bow Pressure” parameter to the same MIDI event that controls the Expression (see “Controller Mapping” section).

For example:

- Expression: CC 11 – Min = 0, Max = 127
- Bow Pressure: CC 11 – Min = 55, Max = 80

Start “from nothing” (fade-in): set the Expression to zero, hit the key with a very low value of the velocity (under MIDI value 10), then increase the Expression as desired.

Jazzy Pizzicato (specific for Double Bass)

Set “Play Mode” to “Pizzicato”, act on “Bow/Pizz Position” and on “Pizz/C.Legno Tone” to find the desired timbre.

If you want a fast decaying sound, decrease the “String Resonance” and “Open Strings” parameters on the “Timbre” page.

To obtain a more wah sound, increase the value of “Pizzicato Fingerboard Interference” on the “Advanced” page.

Usually a Jazzy Pizzicato Double Bass is mono, especially for walking bass style. “Pizzicato Poly” should be set to “Mono Strings Crossing”. You may want to mute any string vibration (i.e. play strictly “mono”) by setting “Mono CrossString Muting” to “All”.

Note-on velocity controls the pluck strength. Very high velocity values lead to a slap sound. Note-off determines the string muting with the same finger used for plucking it.

Open String Pizzicato Strum

The procedure to play an open-string strum is:

“Play Mode” - set to “Pizzicato”

“Pizz./C.Legno Poly” - set to Polyphonic

“Alternate Fingering” - set to "Near the Nut + Open”.

Press Sustain pedal and play the desired open strings, staccato.

Bow Direction Behaviour

The bowing engine determines bow direction according to the phrasing and articulation of the performance.

When a phrase is interrupted by a clear pause, the bow naturally resets to its default direction and position before resuming, according to the set "Bow Start" parameter.

Very short separations between notes (threshold set by the Staccato Interval Time parameter) are interpreted as legato connections, so the bow continues in the same direction. Instead, when staccato notes are played in close succession the bow direction changes with each new attack, though the bow position is not fully reset. Longer stretches of continuous legato playing without a break will eventually force a bow change to preserve a realistic alternation.

Other musical conditions also influence the result. Rapid passages that demand continuous energy, or strongly accented attacks, may trigger a bow change even if the musical line is still connected. This ensures that the simulated bowing remains consistent with natural technique, where alternating directions are required for balance and articulation.

Two special overrides give the performer direct control. Holding the Sustain pedal will force a bow change on every staccato note, even against the normal rules. Conversely, holding the "Keep Bow Direction" KeySwitch (G) will always maintain the same bow direction, regardless of phrasing. This last command has the highest priority in the system. This function can also be mapped to any CC.

This design provides both natural bowing behaviour that follows the musical line and the flexibility for the performer to impose control when needed.

Tips and Troubleshooting

For comprehensive information, FAQs, and more detailed articles, please refer to our Knowledge Base at kb.audiomodeling.com.

No sound

The three most common causes for not getting any sound from a SWAM instrument are:

1. The audio device or audio output may not be correctly assigned. In the Standalone app, navigate to Main Menu -> Settings -> Audio, and select the proper audio device and output channels. If you are using a DAW or host, consult the user manual of the host application.

2. The MIDI device may not be correctly selected. In the Standalone app, navigate to Main Menu -> Settings -> MIDI, and select the proper MIDI device. If you are using a DAW or host, consult the user manual of the host application.

3. The Expression parameter may not have been moved, automated, or controlled by an external MIDI device since the instrument was instantiated.

Ours are expressive instruments. Unlike standard sample libraries, our instruments do not work until the Expression control is changed by an Expression pedal, Breath Controller, wheel, knob, slider, Wind Controller, Seaboard, LinnStrument, Touché, mobile App capable of sending MIDI data, or other MIDI CC11 source.

Can a saxophone emit a sound without blowing into it? Can a violin emit a sound without moving the bow?

Please watch this introductory playlist:

https://youtube.com/playlist?list=PLT7gu1yXoV4HpBDsrDKbVjJKOlhtpsNAI

We provide Factory presets for several MIDI controllers: click "MIDI", click "PRESETS", and choose a preset from the list

The default mapping for the Expression control at startup is the MIDI CC11, which is typically assigned to a keyboard Expression pedal.

The Expression assignment can be customized to match your controller: click "MIDI", click on the Expression Slider, and input the MIDI CC number.

For MIDI keyboards not equipped with an Expression pedal, a good start often is to map Expression to CC1 (Mod Wheel) and Vibrato Depth to AfterTouch (AT).

You can use the "Learn" button and move your controller. In this mode, the instrument recognizes the first incoming CC number and automatically assigns it to the Expression setting.

Even without a physical controller, on a DAW or sequencer you need to draw expression curves (or envelopes) or automations. Please refer to the user manual of your DAW in order to know how to provide MIDI CC curves or automations.

You can find a Getting Started tutorial for most DAWs on our knowledge base https://kb.audiomodeling.com/en/c/how-to

Understanding Ambiente

What is Ambiente and why should I use it?